K-Pop Data Analysis

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Starting in 2024.

Executive Summary

Write something here

Disclaimer

This manuscript is written solely by the author, not by ChatGPT or any other generative AI. We emphasize this due to the widespread presence of AI-created articles.[1] Moreover, the opinions and views expressed in this manuscript are those of the author, and do not necessarily state or reflect those of any institution or government entity.

1 Introduction

Need to add: How to compile this .Rmd into pdf using command line
Need to write a Makefile and the compilation instructions.
Don't just use the RStudio IDE!

Briefly describe what the K-pop music is, such as songs and elements

K-pop music (Korean popular music) is a genre of entertainment originated from South Korea (Lie, 2015), and K-pop emphasizes more on the dancing component than the singing element [citation needed].

Therefore, K-pop shows are like an audio-visual treat, appealing to the general public (Pal and Saha, 2022).

Music: K-pop groups often showcase a wide range of musical styles within a single discography, and common styles include pop, hip-hop, rock, and R&B (Rhythm and Blues) (Liu, 2024).

Lyrics:
Upbeat and catchy - Many tracks are designed with high energy and memorable hooks, making them easy to listen to and popular for dancing [citation needed].

e.g. Debut single song - "Into the New World" (再次重逢的世界)[2]

Performed by *Girls' Generation* (少女時代)[3] in 2007.

[1] https://graphite.io/five-percent/more-articles-are-now-created-by-ai-than-humans
[2] https://en.wikipedia.org/wiki/Into_the_New_World
[3] https://en.wikipedia.org/wiki/Girls%27_Generation

This song is considered a textbook example of K-pop music (Epstein, 2014).

What do you think about the lyrics in this K-pop group?
Positive, motivational, about women's empowerment (Tiensha and Noviana, 2025)

Dance: Choreography, complex

Fashion: Dress, make-up, hair style, etc.

Stage and video effects: Bright lights, video-editing for focus, etc.

Important: Write about why K-pop music is so popular across the globe.

K-pop has emerged popularity worldwide since the early 2010's (Khiun, 2013; Sun, 2022). K-pop's international reputation stems from its versatile music, eye-catching performances, and the strategic use of social media, fostering a global community of dedicated fans who actively promote the artists and genre (Kim and Kwon, 2022; Chen, 2023). Therefore, K-pop enjoys global fame – not only across East Asian countries like China and Thailand (Malik, 2023), but also in the United States[4] and Western Europe[5] (Miroudot, 2024).

In addition to the K-pop performance itself, survival reality shows have also been an emerging trend in the K-pop industry (Butsoontorn, 2023; Lee, 2020). The main objective of such shows is to select the top contestants to form a new K-pop group. One major feature is the audience involvement – many survival shows allow the audience to vote on contestants' performance (Kim and Shin, 2025), keeping the audience engaged with the live episodes.

Need a transition sentence here

Therefore, such TV programs often have high viewership. [citation needed]

For example, Mnet[6] is a television music channel famous for hosting idol-making survival shows, such as *Produce 101* and *Produce 48* (Capistrano and Ramirez, 2023). Their YouTube channel, *Mnet K-POP*,[7] has more than 22 million subscribers worldwide.

Thanks to translation [citation needed], K-pop shows can be enjoyed by people outside the Korean-speaking community.

Do we need to write something more?

1.1 Motivation

The author became interested in K-pop music from the debut of Tzuyu (Chou Tzu-Yu, 周子瑜).[8] Tzuyu is originally from Taiwan, the country in which the author grew up. In 2015, Tzuyu participated in the South Korean reality television show *SIXTEEN*,[9] and eventually got added to the newly-formed girl group *TWICE*.[10] In early 2016, Tzuyu was forced to apologize after she raised the Taiwan flag in a Korean entertainment show (Ahn and Lin, 2019).[11] The flag controversy incident made headline news in Taiwan,[12] and it was estimated to bring in 500,000 votes for the 2016 Taiwan presidential election.[13]

Then in 2017, another Taiwanese girl, Snowbaby (蔡瑞雪),[14] joined the Korean live reality show *Idol School* (偶像學校).[15] At the end of the show, nine winners were selected to form the new girl group *fromis_9*.[16] This

[4] https://bit.ly/4iIHpor
[5] https://nolae.eu/blogs/news/diese-k-pop-konzerte-werden-2025-in-europa-stattfinden
[6] https://en.wikipedia.org/wiki/Mnet_(TV_channel)
[7] http://www.youtube.com/@Mnet
[8] https://en.wikipedia.org/wiki/Tzuyu
[9] https://en.wikipedia.org/wiki/Sixteen_(TV_program)
[10] https://en.wikipedia.org/wiki/Twice
[11] https://bit.ly/3DOcNlP
[12] https://bit.ly/4k5j7ps
[13] https://bit.ly/3CUQWsK
[14] Snowbaby's YouTube channel: https://www.youtube.com/@snowbaby
[15] https://en.wikipedia.org/wiki/Idol_School_(2017_TV_series)
[16] https://en.wikipedia.org/wiki/Fromis_9

also generated much discussion in the Mandarin-speaking community.[17] Although Snowbaby was eliminated in the middle of the show, she still deserves praise for her courage to participate as an international contestant.[18] Since Snowbaby graduated from the same high school as the author did,[19] Snowbaby's experience motivated the author to learn more about K-pop.

The author also followed other K-pop reality shows such as *Produce 48* in 2018.

She loved the visual aesthetics and the artistic performances of K-pop.

To combine with statistical analysis ?!

Explain why and what we are doing in this analysis

Important: Write about the K-pop scandal revealed in 2019 and later.

Although the author enjoyed watching the K-pop survival reality shows, ...

In 2019, the author stopped following the shows produced by Mnet due to the ethical scandals revealed (Miao, 2020), especially the manipulation of voting outcomes on survival shows (Leinonen, 2024).

Therefore, we focus on the ratings directly given by the vocal and dance instructors, rather than the number of audience votes announced by the show organizers.

https://en.wikipedia.org/wiki/Mnet_vote_manipulation_investigation

1.2 K-Pop Controversies

Unfinished below

Behind the beautiful K-pop music and dance videos, ...

Blood and sweat ?! Also money and political capital involved. [citation needed]

A lot more content here, probably need a transition paragraph

Survivorship bias is prevalent in the entertainment industry, and the K-pop genre is no exception (Lockwood, 2021). To become a K-pop idol in Korea, aspiring kids usually start as a trainee at an entertainment company in their early teens (Lee and Jin, 2019). The trainees take vocal and dance lessons, and the training process is intense and extremely competitive (Kang, 2017; Lee, 2024). That's why most K-pop artist performances are polished and well-rehearsed (Kim et al., 2021). Nevertheless, very few trainees can eventually get selected to debut and perform on stage as the company's official group (Han and Pothong, 2021), and even fewer K-pop performers can achieve iconic status (Min, 2024). While a performing group enjoys the spotlight and fame, the group typically lasts only a few years before the contract ends (Cho et al., 2023).

K-pop overcrowded market (Liu, 2025).

Strict body weight and dietary requirements, especially for girl trainees (Venters and Rothenberg, 2023; Zysik, 2024).[20]

Need another transition paragraph

A K-pop vote manipulation scandal was surprisingly revealed in 2019, starting with the *Produce X 101*[21] and the mysterious 29,978 number.[22] After the producer Mnet published the number of votes each contestant received, people calculated the difference between rankings and noticed a strange pattern of the numbers. The difference was exactly 29,978 votes for five intervals among the rankings from 1st to 10th. The pattern

[17] https://www.epochtimes.com/b5/17/7/2/n9346573.htm
[18] https://bit.ly/41p3pym
[19] https://bit.ly/424u3gv
[20] https://www.chinatimes.com/realtimenews/20200928003058-260404?chdtv
[21] https://en.wikipedia.org/wiki/Produce_X_101
[22] https://www.koreaboo.com/news/produce-x-101-rigged-votes-final-members/

seemed to be generated by some mathematical function, and it was nearly impossible to be a coincidence. Hence people suspected that the voting results were manipulated by Mnet,[23] resulting in a lawsuit against Mnet and other companies involved (Choi, 2023).

Finally, Mnet admitted to manipulating the votes in the *Produce 101* series[24] and the subsequent reality shows, including *Idol School*.[25] The show producers rigged the votes in return for financial favors, resulting in not only unfair competition but also employment fraud (Lee and Zhang, 2021). The court demanded that Mnet must provide monetary compensation to the affected trainees, who should have been selected for the debut but lost the opportunity.[26] These entertainment agency representatives were charged with bribery, fraud, and sabotage (Yoshimitsu, 2020).

Idol School: Vote Manipulation Investigation (2019)
https://www.ptt.cc/bbs/KoreaStar/M.1624467107.A.D7F.html

What was the penalty of these entertainment agency representatives?
https://www.ptt.cc/bbs/KoreaStar/M.1680484737.A.28A.html

Need academic citations, not just news links.

Also need to mention other scandals with citations

e.g. sexual harassment (Luu, 2022; Varianna and Kusumawardani, 2024)
e.g. exploitative contracts that mainly benefit the company, not the artists (An, 2025)

Negative impact on the K-pop industry, such as the cancel culture from K-pop fans (Putri and Kusuma, 2025; Febrianti et al., 2023).

Impact on the trainees who lost the chance to debut:
They received monetary compensation, but the debut opportunity could not be recovered. (citation needed)

After the 2019 scandal, Mnet has been under controversy but is still actively producing K-pop dance survival shows.[27] Recent works of Mnet include *Kingdom: Legendary War* (2021)[28] and *Stage Fighter* (2024).[29]

As of 2025: the Deepfake and Generative AI videos pertinent to K-pop. (citation needed)

Deepfake videos of K-pop idols are negatively perceived by the fans (Wang and Kim, 2022a,b), and such malicious use of AI damages not only the idols' reputation but also the whole K-pop industry (Alexandri, 2024).

K-pop performance videos published in 2021 or earlier are mostly real, because they were created before the widespread use of generative AI (Cho and Lee, 2025).

1.3 Technical Narrative

Datasets from K-pop reality shows in South Korea

In this project, we explore a few datasets from K-pop reality shows that have at least two sets of contestant ratings. We would like to evaluate the correlation and consistency between the ratings of the same K-pop trainees, so at least two sets of ratings are necessary. Note that contestant ratings differ from audience votes – the former are given by professional K-pop coaches, while the latter come from the general public.[30]

[23] https://bit.ly/4iWtPh0
[24] https://bit.ly/4oaMEzV
[25] https://www.popdaily.com.tw/korea/846603
[26] https://bit.ly/44gkSKG
[27] https://www.ptt.cc/bbs/KoreaStar/M.1618588754.A.7C9.html
[28] https://en.wikipedia.org/wiki/Kingdom:_Legendary_War
[29] https://en.wikipedia.org/wiki/Stage_Fighter
[30] For Mnet-hosted shows, voting requires an account on the Mnet Plus app. https://www.mnetplus.world/en/

Unfinished below

Datasets Considered: Need to have at least two sets of contestant ratings to evaluate for correlation and/or consistency.

- *Idol School* (2017)
  - Female contestants
  - Basic Strength Ratings: Vocal, Dance, Physical, Overall
  - https://en.wikipedia.org/wiki/Idol_School_(2017_TV_series)
  - https://en.wikipedia.org/wiki/List_of_Idol_School_contestants
- *Produce 101* (2016)
  - Female contestants
  - Judges' Evaluation 1: A, B, C, D, F
  - Judges' Evaluation 2: A, B, C, D, F
  - https://en.wikipedia.org/wiki/Produce_101
  - https://en.wikipedia.org/wiki/Produce_101_season_1
  - https://en.wikipedia.org/wiki/List_of_Produce_101_contestants
- *Produce 101 Season 2* (2017)
  - Male contestants
  - Judges' Evaluation 1: A, B, C, D, F
  - Judges' Evaluation 2: A, B, C, D, F
  - https://en.wikipedia.org/wiki/Produce_101_season_2
  - https://en.wikipedia.org/wiki/List_of_Produce_101_contestants_(season_2)
- *Produce 48* (2018)
  - Female contestants (50% from South Korea and 50% from Japan)
  - Judges' Evaluation 1: A, B, C, D, F
  - Judges' Evaluation 2: A, B, C, D, F
  - https://en.wikipedia.org/wiki/Produce_48
  - https://en.wikipedia.org/wiki/List_of_Produce_48_contestants
- *Produce X 101* (2019)
  - Male contestants
  - Judges' Evaluation 1: A, B, C, D, X
  - Judges' Evaluation 2: A, B, C, D, F, X
  - https://en.wikipedia.org/wiki/Produce_X_101
  - https://en.wikipedia.org/wiki/List_of_Produce_X_101_contestants

Datasets NOT Considered: These datasets do not have enough sets of contestant ratings for our analysis.

For instance, *Mix Nine* (2017-2018)[31] has only one set of contestant ratings at the beginning of the show.

In fact, many K-pop survival shows do not provide even one set of contestant ratings. One example is *SIXTEEN* (2015),[32] which we mentioned earlier in Section 1.1. Other examples include *I-Land* (2020),[33] *Girls Planet 999* (2021),[34] and *Street Man Fighter* (2022).[35]

This manuscript is created using `R` Markdown (Allaire et al., 2024)[36] for reproducible data analysis, just like our earlier technical report about the education in Taiwan (Chai, 2024). We have posted our code and data on GitHub,[37] so readers can download the GitHub repository and play with the script themselves.

[31] https://en.wikipedia.org/wiki/Mix_Nine
[32] https://en.wikipedia.org/wiki/Sixteen_(TV_program)
[33] https://en.wikipedia.org/wiki/I-Land
[34] https://en.wikipedia.org/wiki/Girls_Planet_999
[35] https://en.wikipedia.org/wiki/Street_Man_Fighter
[36] https://rmarkdown.rstudio.com/
[37] https://github.com/star1327p/K-Pop-Dataset

The rest of this manuscript is organized as follows.

e.g. Chapter 23 does something.

2 *Idol School* Dataset (2017)

Idol School (偶像學校) is a 2017 Korean reality television show produced by Mnet, selecting nine winners were selected to form the new girl group *fromis_9* in 2018.[38] Unlike the 2016 *Produce 101* that mostly featured K-pop trainees,[39] *Idol School* did not require any vocal or dance experience and was willing to train the participants from scratch. Despite the low barrier to entry, many participants in the reality show had previously trained under various entertainment companies.[40] For example, NATTY was trained under JYP Entertainment[41] and made it to the finals of the *SIXTEEN* reality show[42] in 2015. Lee Yoo Jeong (李悠汀) previously debuted in the girl group *myB*, but the group disbanded in 2016.[43]

For *Idol School*, we need to add some academic citations.

On the other hand, some contestants quickly learned to perform K-pop at a high level, showing that it's possible to succeed with little-to-no initial experience. For example, Lee Si An (李詩安) and Yoo Ji Na (柳知娜) started out as complete beginners in the show, but they picked up the K-pop dance very fast and survived until the final episode. Baek Ji Heon (白知憲) and Lee Na Gyung (李娜景) also scored low in the initial performance test, but both of them eventually made it to the top nine for the debut.

Unfortunately, Snowbaby (蔡瑞雪) did not do well and got eliminated in the first round (fourth episode of the series). Snowbaby had not received any vocal or dance training in K-pop, so she was a complete beginner in the show. Moreover, Korean is not Snowbaby's native language, making the *Idol School* training extremely tough for her. Despite the major setbacks, Snowbaby still cherished the *Idol School* experience and was willing to publicly share what she learned during the K-pop bootcamp.[44]

Need another transition sentence or paragraph

In the live reality show *Idol School*, nine winners were selected to form the new girl group *fromis_9*.[45] This girl group debuted in 2018 and remained active until the contract with Pledis Entertainment ended in 2024. In January 2025, five members of the group signed a new contract with ASND.[46]

What was the announced selection process for the nine winners?
Audience voting
What was the actual selection process instead?

Need to write the data description

Wikipedia data: https://en.wikipedia.org/wiki/List_of_Idol_School_contestants

2.1 Read in the *Idol School* Dataset

We manually copy-pasted the contestant data from Wikipedia into a Microsoft Excel workbook (`.xlsx`), and used the `R` package `readxl` (Wickham and Bryan, 2023) to load the dataset. A main advantage of `.xlsx` over `.csv` is that we can have multiple data sheets in the same Excel file for consolidation. Moreover, Excel supports Chinese characters, so we can also include the Chinese names of each contestant. Since the English

[38] https://en.wikipedia.org/wiki/Fromis_9
[39] https://en.wikipedia.org/wiki/Produce_101
[40] https://kpop.fandom.com/wiki/Idol_School
[41] https://en.wikipedia.org/wiki/JYP_Entertainment
[42] https://en.wikipedia.org/wiki/Sixteen_(TV_program)
[43] https://zh.wikipedia.org/wiki/MyB
[44] Snowbaby's sharing on YouTube: https://www.youtube.com/watch?v=WMUBDbNsuGU
[45] https://en.wikipedia.org/wiki/Fromis_9
[46] https://kpop.fandom.com/wiki/ASND

translation of Korean names look similar to each other (Kim, 2020), we also include the date of birth (DOB) to make it easier to uniquely identify each contestant. For those who are able to read Chinese, we put each contestant's name in Chinese characters as well.

Need to briefly describe the data and the initial test in the show

Specify the column names we included, also the column names we printed here.

For each contestant, we focus on these columns below:

- **Name_Chn**: Chinese name of the contestant.
- **Name_Eng**: English name of the contestant.
- **DOB**: Date of birth (yyyy-mm-dd) of the contestant.[47]
- **Vocal**: Vocal score in the initial test, ranging from 0 to 10 in 0.1 point increments.
- **Dance**: Dance score in the initial test, ranging from 0 to 10 in 0.1 point increments.
- **Physical**: Physical score in the initial test, ranging from 0 to 10 in 0.1 point increments.
- **Overall**: Average of the vocal, dance, and physical scores.

Additional variables:

- **Ability_Rank**: Rank of the contestant by the initial test score average.
- **Final_Rank**: Final rank of the contestant published by the show.
- **Round_Eliminated**:
  - R1: Eliminated in the first round.
  - R2: Eliminated in the second round.
  - R3: Eliminated in the third and penultimate round.
  - R4: Eliminated in the fourth and final round.
  - Survived: Selected for the debut and joined the newly-formed K-pop group.
  - X: Left the show before the first round of elimination started.
- **Special_Notes**: Notes that are worth mentioning but do not belong to any other column, such as nationality (outside South Korea) and score error correction.

Add the metadata in the Excel file or the Appendix ?!

Currently I prefer adding the metadata in the Excel file for proximity to the data itself.

Show the first 10 records as a snapshot of the dataset.

```r
library(readxl)
idol_school = read_excel("UNFINISHED_Idol_School_Dataset.xlsx",
                         sheet="Idol_School_Dataset")

# Date of birth (DOB) should be date only, not a full timestamp.
idol_school$DOB = as.Date(idol_school$DOB)

columns_to_show = c("Name_Chn", "Name_Eng", "DOB",
                    "Vocal", "Dance", "Physical", "Overall")

idol_school[1:10, columns_to_show]
```

```
## # A tibble: 10 x 7
##    Name_Chn Name_Eng       DOB        Vocal Dance Physical Overall
```

[47] In the Excel file, the date of birth is shown in m/d/yyyy format.

```
##    <chr>    <chr>          <date>     <dbl> <dbl>    <dbl>   <dbl>
##  1 NATTY    NATTY          2002-05-30   9.8   8          8.1    8.63
##  2 劉怡伶   Tasha          1993-10-11   8     9.5        8      8.5
##  3 李采映   Lee Chae Young 2000-05-14   8.5   8.5        7.5    8.17
##  4 宋河英   Song Ha Young  1997-09-29   8.6   5.9        9.8    8.1
##  5 金恩書   Kim Eun Suh    2000-11-14   6.3   6.9       10      7.73
##  6 金明智   Kim Myong Ji   1997-10-09   5.5   7.9        8.2    7.2
##  7 張圭悧   Jang Gyuri     1997-12-27   7.2   7.1        7      7.1
##  8 朴宣     Park Sun       2004-05-25   9.5   6.1        5.5    7.03
##  9 李悠汀   Lee Yoo Jeong  1997-02-26   5.8   6.2        9      7
## 10 金娜妍   Kim Na Yeon    1996-05-15   8.3   6          6.4    6.9
```

A total of 41 contestants entered the *Idol School* reality show. But the 41st contestant, Som Hye In (慎惠仁), left the show early due to health reasons. She was unable to complete the basic test, so her score was zero in all three categories (vocal, dance, and physical). Hence we remove the last row (all zeros) and keep only the remaining 40 contestants' scores in the dataset.

```
# We MUST remove the 41st contestant's scores (all zeros).
idol_school = idol_school[1:40,]
```

2.2 *Idol School*: Exploratory Data Analysis

Context: Write about how the vocal, dance, and physical scores were evaluated.

Vocal testing

Mostly about the singing techniques, rather than memorizing the Korean lyrics.

Dance testing

Physical testing contains a group exercise and an individual exercise.

Also mention the top performers in overall scores and in each category.

2.2.1 Error Correction in Data

To observe the patterns of the *Idol School* data published on Wikipedia,[48] we first sorted the scores of each category (vocal, dance, and physical) in descending order. Our presumption is that in each category, no two contestants should have the same score. However, after sorting the *Idol School* data, we found two 3.5's and two 1.2's in the physical scores. Especially that the two 3.5's belong to top-ranked contestants Bae Eun Yeong (裴恩英) and Park Ji Won (朴池原), this issue quickly caught our attention to make corrections to the data. Fortunately, the *Idol School* show's videos are available on the Bilibili platform, with subtitles in Simplified Chinese that we can understand.[49] We saved the screenshots from the contestant scores in the first episode.[50] Hence we can compare the Wikipedia data with the exact scores published in the show.

In the video clip, Park Ji Won (朴池原) and her partner were the first runner-up in the group physical exercise.[51] We are surprised that Ji Won's physical score was only 3.5. According to the video's score table for contestants ranked 11th to 20th,[52] Ji Won's physical score should be 6.2. The Wikipedia table shows an inconsistency in Ji Won's overall score, i.e., the average across the three categories. Ji Won's vocal score was 7.9, and her dance score was 5. These numbers seem to be reasonable for Ji Won, because she is known for

[48] https://en.wikipedia.org/wiki/List_of_Idol_School_contestants#Elimination_chart
[49] https://www.bilibili.com/video/BV1554y1C7wj/
[50] Screenshots saved: https://github.com/star1327p/K-Pop-Dataset/tree/main/Idol_School_Rating_Screenshots
[51] Screenshot of the group physical exercise: https://bit.ly/4a7QT9m
[52] https://bit.ly/400KUhH

excellent singing and good dancing as a performer.[53] Therefore, we assume both scores to be correct. If the physical score had really been 3.5, then Ji Won's overall score would be 5.47, dropping her from 13th place to the 18th. If the overall score of 6.37 had been correct, then Ji Won's physical score should be 6.2. The second scenario is more likely to be true, given the evidence we found in the video clip. Hence we corrected Ji Won's physical score to 6.2.

The two 1.2 physical scores are more difficult to check for the underlying values, probably because they occurred in two contestants of lower ranking.[54] The two contestants, Jessica Lee (李瑟) and Michelle White (懷特·米雪兒), ranked in the lower half of all 41 contestants in terms of the overall ability test. Both of them got eliminated in the first round, so they did not receive much attention in the show. With the help of Google Translate,[55] we were able to translate the image of Korean text to (readable) English. Finally, we discovered that Michelle White's physical score should be 1.3, not 1.2.

2.2.2 Summary Statistics

Check for the mean and median of each category score

Let's check for the mean and median of each category score. Wait - we can do better by obtaining the summary statistics. The five-number summary refers to the five most important percentiles in the data sample – minimum, 1st quartile (25th percentile), median (50th percentile), 3rd quartile (75th percentile), and maximum.[56] The `summary` function in `R` outputs the five-number summary along with the arithmetic mean for the data.

```r
# Combine all three summary tables
vocal_summary = summary(idol_school$Vocal)
dance_summary = summary(idol_school$Dance)
physical_summary = summary(idol_school$Physical)

score_summary = rbind(vocal_summary, dance_summary, physical_summary)
row.names(score_summary) = c("Vocal","Dance","Physical")
print(score_summary)
```

```
##          Min. 1st Qu. Median   Mean 3rd Qu. Max.
## Vocal     1.0   2.875   4.95 4.8850   6.425  9.8
## Dance     1.0   3.825   5.55 5.4850   7.025  9.5
## Physical  0.4   1.675   3.25 4.1925   6.425 10.0
```

If the readers prefer a graphical summary of the data distributions, Figure 1 is the **box plot** of the vocal, dance, and physical scores in the *Idol School* show. We visualize the data using `ggplot2` (Wickham, 2016) to compare the three sets of scores. For each category, the central bold line indicates the median score. The box ranges from the 1st quartile to the 3rd quartile. The whiskers represent the minimum and maximum values in the sample.[57]

Alternative: Hide the code and only show the graph itself

[53] Park Ji Won was the main vocalist in *fromis_9*. https://bit.ly/402yCFI
[54] Physical scores of all contestants in *Idol School*: https://bit.ly/3DRNK0Z
[55] https://translate.google.com/
[56] http://en.wikipedia.org/wiki/Five-number_summary
[57] It is possible to have outliers beyond the upper and lower whiskers. https://bit.ly/3LSPTxu

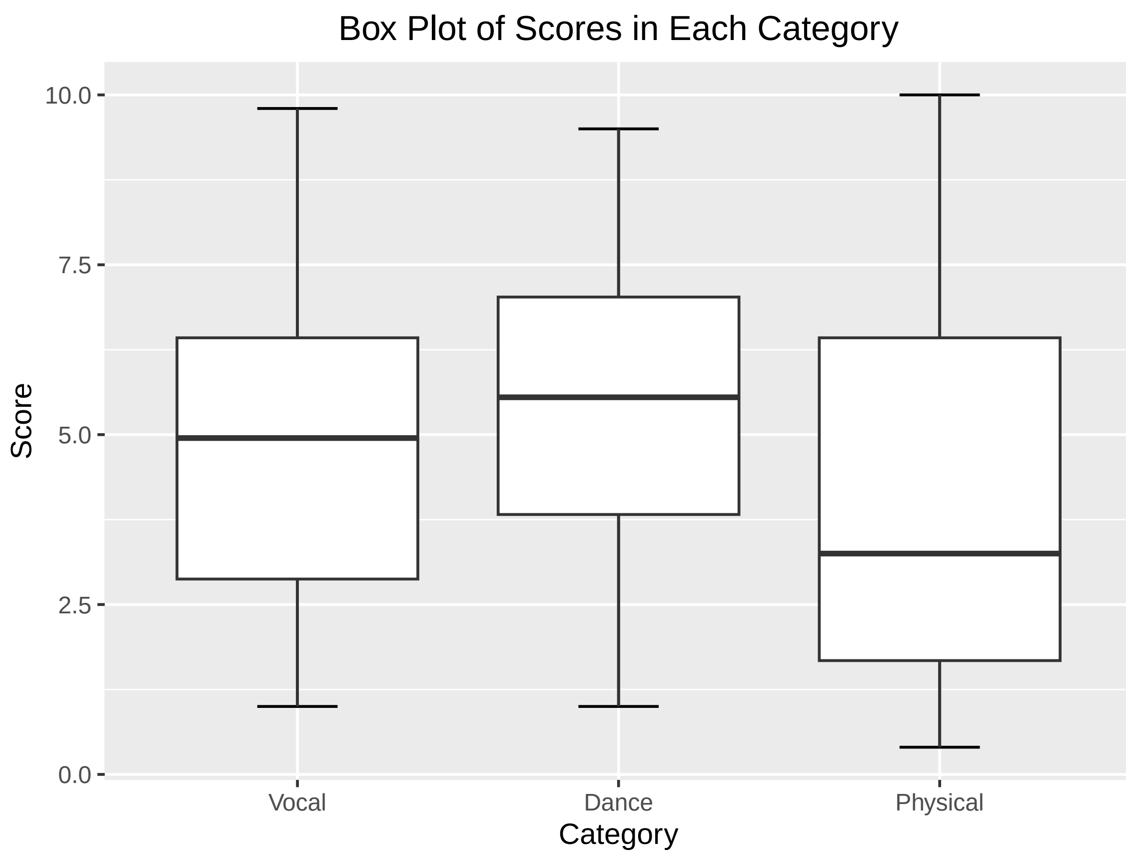

Figure 1: Box plot of the vocal, dance, and physical scores in the *Idol School* show

Observe that vocal and dance scores have 1 as the minimum score (who participated but made a blunder while performing), and the maximum score is below 10 (full mark). Physical scores have a wider range because the first place is automatically given a 10, and the other contestants' scores are calculated using the best performer as the baseline. Hence it is possible to receive a physical score below 1.

Vocal has a median score of 4.95 (rounding to 5), which indicates decent performance. NATTY received 9.8 as the highest vocal score, and the instructor said her singing performance was perfect (considering that NATTY is originally from Thailand and hence not a native speaker in Korean).[58] On the other hand, Lee Hae In (李海印) received a score of 1 in vocal, because she unfortunately lost her voice and could barely sing at that time.

Dance:

Dance has the highest median score (5.55) among the three categories because … (elaborate on the reasons)

Although the best score was only 9.5 (given to Tasha (劉怡伶)), the median score of dance is the highest (5.55) among the three categories.

In K-pop, dance is an essential element for performers (Lee, 2018; Auh, 2025). In the live band, everyone must dance!

K-pop trainees: More talent in dance than in vocal (singing). (citation needed)

The video clip showed that Lee Si An (李詩安) and Yoo Ji Na (柳知娜) struggled in the dance exercise, but it was Jung So Mi (鄭昭彌) who got the lowest score of 1.

[58]NATTY's birth name is Anatchaya Suputhipong. https://en.wikipedia.org/wiki/Natty_(Thai_singer)

c.f. Vocal roles are divided into main vocal, lead vocal, and sub-vocal (i.e., everyone else, or simply "vocal").[59]

Generally the most challenging lines are assigned to the main vocal and the lead vocal, while the other vocals receive easier parts. (citation needed)

Physical:

Kim Eun Suh (金恩書) came in first and received the full mark of 10, while Song Ha Young (宋河英) got 9.8 as the second place.

The physical scores are harsher than the vocal and dance scores. In the vocal and dance activities, the lowest score is 1 to account for the contestant's participation. However, the lowest physical score observed is 0.4, and a total of five students received a score below 1. Physical also has the lowest median score (3.25) among all three categories.

The show did not reveal how the physical scores were calculated from the raw time sustained by each participant in each exercise.

Physical: group physical exercise vs individual exercise

But after watching the video clip on the physical tests, we suspect that the individual exercise accounted for a larger proportion of the physical score. We saw in the video that Lee Da Hee (李多熙) and Kim Na Yeon (金娜妍) came in first in the group physical exercise. However, Da Hee's physical score was only 4.9, and Na Yeon got 6.4 (which is good but not one of the best).[60]

Remark: That's why it is important to remove the record with all zeros, otherwise the minimum score would be zero in every single category. Then we would not have noticed that the vocal and dance scores start from 1 with mere partipation, but the physical scores have a different scale.

Comparison:

The median in physical score (3.25) is lower than the median in vocal (4.95) or dance (5.55), indicating that many contestants did not do well in the physical test. In fact, some contestants had a remarkably high score in dance but a low score in physical. One example is Bae Eun Yeong (裴恩英), who scored 9.3 in dance (second place) and 3.5 in physical (slightly above the median of 3.25). Another example is Lee Hae In (李海印), who got 8.4 in dance but only 1.8 in physical.

With a few exceptions:

e.g. Jo Yuri (曹柔理) received only 2.2 in dance but 5.9 in physical. Jo Yuri had not received any K-pop training prior to the show in 2017, but she was a long jump athlete during her school years.[61]

e.g. Song Ha Young (宋河英) got 5.9 in dance and 9.8 in physical. Song Ha Young was a certified aerial yoga instructor before she first appeared in the *Idol School* reality show.[62]

2.2.3 Correlation between Vocal, Dance, and Physical Scores

We can obtain the pairwise correlation coefficients of each category. There is a positive association between the vocal, dance, and physical scores. We round each number to three decimal places.

```
vocal_vs_dance = round(cor(idol_school$Vocal, idol_school$Dance), 3)
dance_vs_physical = round(cor(idol_school$Dance, idol_school$Physical), 3)
vocal_vs_physical = round(cor(idol_school$Vocal, idol_school$Physical), 3)

# Use the cat function to output multiple lines at a time
cat(paste0("Correlation of vocal and dance:    ", vocal_vs_dance, "\n",
```

59 https://bit.ly/3RFAHDL
60 Screenshot of the first place team in the group physical exercise: https://bit.ly/3EDBhz5
61 https://bit.ly/431HHRS
62 https://bit.ly/3ZJQipK

```
            "Correlation of dance and physical: ", dance_vs_physical, "\n",
            "Correlation of vocal and physical: ", vocal_vs_physical))
```

```
## Correlation of vocal and dance:    0.645
## Correlation of dance and physical: 0.509
## Correlation of vocal and physical: 0.664
```

Alternatively, we can also compute the correlation matrix for the three score categories (variables). The diagonal elements are always exactly 1 – because they represent the correlation of a variable with itself, which is a perfect positive correlation. The off-diagonal elements indicate the correlation coefficient between different categories.

```
round(cor(idol_school[,c("Vocal","Dance","Physical")]), 3)
```

```
##          Vocal Dance Physical
## Vocal    1.000 0.645    0.664
## Dance    0.645 1.000    0.509
## Physical 0.664 0.509    1.000
```

Create the scatterplots and/or correlation plots using `ggplot`!

Figure 2 contains the three scatterplots of the pairwise correlation.

Vocal vs dance has the highest linear correlation, with many points on the diagonal line.

Vocal vs physical also has a high correlation, although the graph seems slightly more spread-out than vocal vs dance.

Physical vs dance shows a funnel shape in the scatter plot, implying a relatively large variance of physical scores among the participants with high dance scores.

Alternative: Hide the code and only show the graph itself

Idea: Check out physical vs dance – conditional on dance < 5.0 vs dance > 5.0

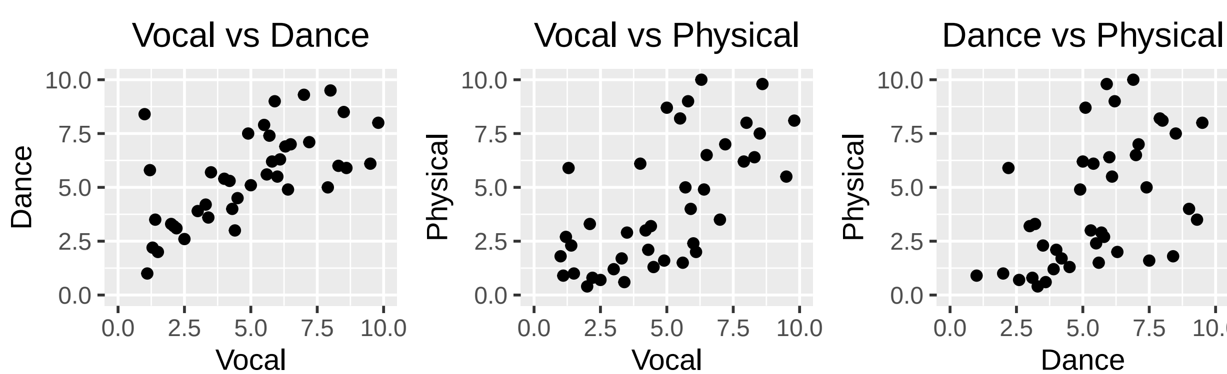

Figure 2: Scatterplots to show the pairwise correlation between the vocal, dance, and physical scores. The graphs from left to right are: Vocal vs Dance, Vocal vs Physical, and Dance vs Physical.

Need to explain the correlation coefficients and the K-pop context.

The training at a K-pop entertainment company in Korea usually includes vocal and dance lessons (Padget, 2017), so it is reasonable to see a high correlation between vocal and dance scores. Theoretically dance and physical should be highly correlated (Ngo et al., 2024), but in the Idol School dataset, we observed a slightly lower correlation in dance vs physical than in dance vs vocal. Physical strength is essential to dancing, but dance also includes other critical elements such as technique and aesthetic expression (Geukes et al., 2023).

Do more analysis to the Idol School data!

Idea: Check out physical vs dance – conditional on dance < 5.0 vs dance >= 5.0

```
# UNFINISHED HERE
dance_upper = idol_school[which(idol_school$Dance >= 5),]
dance_lower = idol_school[which(idol_school$Dance < 5),]

# Correlation coefficients
# cor(dance_upper$Dance, dance_upper$Physical)  # 0.06372215
# cor(dance_lower$Dance, dance_lower$Physical) # 0.1262473

# Standard deviation
# sd(dance_upper$Physical) # 2.773007
# sd(dance_lower$Physical) # 1.645861
```

2.3 *Idol School*: Additional Resources

For the contestants eliminated from the show, some continued to try in K-pop reality shows to earn another position in debut, while some others left the K-pop industry to pursue other goals.[63]

In 2017, a Taiwanese blogger followed the whole *Idol School* series, and used algorithms to predict the final ranking.[64] Both random forests and SVM[65] predicted Lee Hae In (李海印) to be in the top 9 for the debut, but Hae In ended up in an unexpected 11th place.

K-pop fans crowdsourced to count the votes for Hae In, and the number of confirmed votes was already higher than the number of votes published by Mnet. (Note that the number of confirmed votes is a subset of all actual votes.) This raised suspicion that the *Idol School* results may have been rigged.[66]

Then in 2019, the court's investigation revealed widespread manipulation of the participant rankings. (citation needed)

A lot more evidence revealed that the rankings were manipulated by the organizers.

In the final episode of *Idol School*, Lee Hae In (李海印) actually came in first place. However, Mnet adjusted her ranking to 11th, resulting in Hae In being unfairly eliminated.

Elaborate more on the vote manipulation

Idol School: Vote Manipulation Investigation (2019)
https://www.ptt.cc/bbs/KoreaStar/M.1624467107.A.D7F.html

Consider creating a table to document the number of votes manipulated

Columns:

- Episodes 1, 2, …, 11
- Number of participants
- Number of rankings that were fabricated

[63] https://www.ptt.cc/bbs/fromis_9/M.1555819461.A.C73.html
[64] https://shavid.pixnet.net/blog/post/331691281
[65] SVM stands for Support Vector Machine. https://en.wikipedia.org/wiki/Support_vector_machine
[66] https://star.ettoday.net/news/1022482

```r
# UNFINISHED HERE
# Consider: Hide the code and only show the table itself
# Use: echo=FALSE

episode = c("1st","2nd","3rd","4th","5th",
            "6th","7th","8th","9th","10th","11th")
participants = c(41,40,40,40,32,32,28,28,28,18,18)
fabricated   = c( 0,33, 0,39,26,27,27,27,26,17,17)

evidence_df = data.frame(episode, participants, fabricated)
names(evidence_df) = c("Episode", "Participants", "Fabricated Rankings")

# Add a new column "Notes" to write more comments
evidence_df$Notes = c(rep(" ",11)) # initialize each row to blank
evidence_df$Notes[1] = "No evidence of fabrication found."
evidence_df$Notes[3] = "No rankings published."
evidence_df$Notes[4] = "R1: Moved 3 people to survival."
evidence_df$Notes[6] = "R2: Moved 2 people to survival."
evidence_df$Notes[9] = "R3: Moved 2 people to survival."
evidence_df$Notes[11]= "R4: Moved 3 people to the debut."
names(evidence_df)[4] = "Special Comments or Notes"

# Add a Total row at the bottom of the data.frame
total_participants = sum(participants) # 345
total_fabricated = sum(fabricated) # 239
ratio_fabricated = round(total_fabricated/total_participants, 3) # 0.693
# 239/345 ~= 69.3%

final_row = c("Total", total_participants, total_fabricated,
              paste0("A total of ",ratio_fabricated*100,"% were fabricated."))

evidence_df = rbind(evidence_df, final_row)
evidence_df
```

```
##    Episode Participants Fabricated Rankings         Special Comments or Notes
## 1      1st           41                   0 No evidence of fabrication found.
## 2      2nd           40                  33
## 3      3rd           40                   0            No rankings published.
## 4      4th           40                  39   R1: Moved 3 people to survival.
## 5      5th           32                  26
## 6      6th           32                  27   R2: Moved 2 people to survival.
## 7      7th           28                  27
## 8      8th           28                  27
## 9      9th           28                  26   R3: Moved 2 people to survival.
## 10    10th           18                  17
## 11    11th           18                  17  R4: Moved 3 people to the debut.
## 12   Total          345                 239 A total of 69.3% were fabricated.
```

Nearly 70% of the rankings were artificially modified by the show organizers.

In other words, the results were pre-determined before the survival show had even started!

Due to the manipulation of votes, it is meaningless to predict the final rankings in the *Idol School* survival show. That's why we decided to analyze only the contestant scores given by the professional instructors.

These scores are a relatively good benchmark of raw performance, especially that they were not artificially changed to favor any particular contestant.

3 *Produce 48* Dataset (2018)

Produce 48 dataset (2018)

Wikipedia data: https://en.wikipedia.org/wiki/Produce_48

Need to write the background information of the reality show

Produce 48 featured 96 contestants primarily from South Korea and Japan. The show was a collaboration between the Mnet's *Produce 101* series[67] and the Japanese *AKB48* idol group (López Del Valle, 2021).[68]

Write a sentence about the prevalence of multiculural K-pop reality shows (citation needed)

Some former contestants in *Idol School* tried again in the *Produce 48* reality show in 2018.

A total of 12 contestants were eventually selected from *Produce 48* to create the time-limited girl group *IZ*ONE*,[69] which was active during 2018-2021 in both Korea and Japan.

3.1 Read in the *Produce 48* Dataset

Produce 48 dataset (2018)

Wikipedia data: https://en.wikipedia.org/wiki/Produce_48

Need to write the data description

Similar data format to the *Idol School* dataset described in Section 2.1.

Same definition as in the *Idol School* dataset:
Name_Chn, **Name_Eng**, **DOB**, **Final_Rank**, **Round_Eliminated**, and **Special_Notes**.

Since there are only 96 participants, contestants who voluntarily left the show early (not due to elimination) are assigned to a 100 in **Final_Rank**.

List the newly-introduced variables here:

- Unfinished here.
- **First_Eval**: Rating of the first evaluation (A, B, C, D, F).
- **Second_Eval**: Rating of the second evaluation (A, B, C, D, F).
  - X: Left the show before completing the second evaluation.
- **Country**: Nationality, most contestants are from Korea or Japan.

Snapshot of the *Produce 48* dataset:

```r
library(readxl)
produce_48_data = read_excel("UNFINISHED_Idol_School_Dataset.xlsx",
                             sheet="Produce_48_Dataset")

# Date of birth (DOB) should be date only, not a full timestamp.
produce_48_data$DOB = as.Date(produce_48_data$DOB)
```

[67] https://en.wikipedia.org/wiki/Produce_101
[68] https://en.wikipedia.org/wiki/AKB48
[69] https://en.wikipedia.org/wiki/Iz*One

```r
columns_to_show = c("Name_Chn", "Name_Eng", "DOB",
                    "First_Eval", "Second_Eval", "Final_Rank")

produce_48_data[1:20, columns_to_show]
```

```
## # A tibble: 20 x 6
##    Name_Chn Name_Eng        DOB        First_Eval Second_Eval Final_Rank
##    <chr>    <chr>           <date>     <chr>      <chr>            <dbl>
##  1 張員瑛   Jang Won Young  2004-08-31 B          B                    1
##  2 宮脇咲良 Miyawaki Sakura 1998-03-19 A          A                    2
##  3 曹柔理   Jo Yuri         2001-10-22 A          F                    3
##  4 崔叡娜   Choi Ye Na      1999-09-29 A          B                    4
##  5 安俞真   An Yu Jin       2003-09-01 B          A                    5
##  6 矢吹奈子 Yabuki Nako     2001-06-18 F          A                    6
##  7 權恩妃   Kwon Eun Bi     1995-09-27 A          C                    7
##  8 姜惠元   Kang Hye Won    1999-07-05 F          F                    8
##  9 本田仁美 Honda Hitomi    2001-10-06 C          A                    9
## 10 金采源   Kim Chae Won    2000-08-01 B          B                   10
## 11 金玟周   Kim Min Ju      2001-02-05 D          C                   11
## 12 李彩演   Lee Chae Yeon   2000-01-11 A          A                   12
## 13 韓霄瑗   Han Cho Won     2002-09-16 D          B                   13
## 14 李佳恩   Lee Ka Eun      1994-08-20 A          A                   14
## 15 宮崎美穂 Miyazaki Miho   1993-07-30 D          D                   15
## 16 高橋朱里 Takahashi Juri  1997-10-03 B          A                   16
## 17 竹内美宥 Takeuchi Miyu   1996-01-12 A          B                   17
## 18 下尾美羽 Shitao Miu      2001-04-03 D          D                   18
## 19 朴海允   Park Hae Yoon   1996-01-10 A          D                   19
## 20 白間美瑠 Shiroma Miru    1997-10-14 B          D                   20
```

Data entry complete for all contestants in *Produce 48*, including those who left in the middle of the show.

Create a matrix for the two sets of ratings.

For each rating, also check how many contestants are from Korea and how many are from Japan.

```r
# UNFINISHED HERE
produce_48_data[81:96, columns_to_show]
```

```
## # A tibble: 16 x 6
##    Name_Chn   Name_Eng        DOB        First_Eval Second_Eval Final_Rank
##    <chr>      <chr>           <date>     <chr>      <chr>            <dbl>
##  1 克利絲汀   Alex Christine  1996-12-09 B          C                   82
##  2 栗原紗英   Kurihara Sae    1996-06-20 F          D                   83
##  3 趙英燕     Cho Yeong In    2001-10-31 B          C                   84
##  4 淺井裕華   Asai Yuuka      2003-11-10 F          D                   85
##  5 安藝媛     Ahn Ye Won      2001-02-10 F          F                   86
##  6 內木志     Naiki Kokoro    1997-04-06 D          C                   87
##  7 金有彬     Kim Yu Bin      2003-02-27 B          D                   88
##  8 趙思朗     Cho Sa Rang     2003-09-05 B          F                   89
##  9 崔韶恩     Choi So Eun     2001-09-19 B          C                   90
## 10 篠崎彩奈   Shinozaki Ayana 1996-01-08 F          F                   91
## 11 元書妍     Won Seo Yeon    2000-05-23 C          F                   92
## 12 月足天音   Tsukiashi Amane 1999-10-26 F          F                  100
```

```
## 13 田中美久    Tanaka Miku     2001-09-12 F          C                  100
## 14 梅山戀和    Umeyama Kokona  2003-08-07 F          X                  100
## 15 植村梓      Uemura Azusa    1999-02-04 F          X                  100
## 16 松井珠理奈  Matsui Jurina   1997-03-08 B          B                  100
```

Let's look at the nationality breakdown of *Produce 48* contestants. Although *Produce 48* advertised a collaboration between Korean and Japanese entertainment groups, the Korea-Japan split is not 1-1 among participants. The majority (56%) of contestants are domestic within South Korea, while a lower but remarkable (40%) proportion is from Japan. Also, two contestants are from China and another one is from the United States.

```r
table(produce_48_data$Country)
```

```
##
## China Japan Korea   USA
##     2    39    54     1
```

3.2 *Produce 48*: Two Evaluations of Contestant Performance

Write some narrative about Produce 48

At the beginning of the show, there were two evaluations to the 96 contestants' talents. Each evaluation involved a sing-and-dance performance and resulted in a letter grade (A-F). Both letter grades were recorded for each contestant.

In the first evaluation, the contestants were required to perform a popular K-pop song as their initial practice. Then the mentors gave each individual a grade based on their performance, and assigned them to temporary training classes at their level.

```r
table(produce_48_data$First_Eval, dnn="First_Eval")
```

```
## First_Eval
##  A  B  C  D  F
## 15 25 22 15 19
```

The second evaluation was to have each contestant perform the *Produce 48*'s theme song "Nekkoya (Pick Me)".[70] After the song was announced, the contestants were given three days to prepare for the choreography and memorize the lyrics. The song has a Korean version and a Japanese version, so each student may choose to perform in their preferred language. Then the students were given their new grades, and reassigned to their new practice classes. The rating outcomes are still A-F, the same as in the first evaluation. The "X" rating means that the contestant left the show before completing the second evaluation.

```r
table(produce_48_data$Second_Eval, dnn="Second_Eval")
```

```
## Second_Eval
##  A  B  C  D  F  X
## 14 20 22 16 22  2
```

[70]https://en.wikipedia.org/wiki/Nekkoya_(Pick_Me)

Cross-table: **First__Eval** as row, and **Country** as column

Let's examine how Korean and Japanese participants scored in the first evaluation. We noticed that the majority of contestants from Japan were placed in D or F (lowest grades), while very few contestants from Korea did. Although the Korean K-pop coaches criticized the Japanese contestants for their performance, we would like to emphasize that the Idol training process differs greatly in Korea and Japan.[71] In Korea, the primary focus is on vocal and dance techniques, while in Japan, the Idol training values artistic interpretation and individual expressiveness (Lee, 2023).

In the show, many Japanese contestants bursted into tears due to the harsh feedback given by the Korean K-pop instructors.[72] The Japanese contestants enjoyed highly positive responses in their own country, so they thought they would receive an A in the evaluation during *Produce 48*. Instead, most of them received an F (failure). After the first evaluation, some of the Japanese contestants withdrew from the show for various reasons. Two of them did not even complete the second evaluation.[73]

```
table(produce_48_data$First_Eval, produce_48_data$Country,
      dnn=c("First_Eval","Country"))
```

```
##           Country
## First_Eval China Japan Korea USA
##          A     0     2    13   0
##          B     2     4    18   1
##          C     0     5    17   0
##          D     0    11     4   0
##          F     0    17     2   0
```

Cross-table: **Second__Eval** as row, and **Country** as column

What about the second evaluation?

Observation: Contestants from Japan faced a little better in the second evaluation – approximately half of them received a satisfactory grade (A, B, or C).

The scoring of domestic contestants in Korea became harsher this time – one third of them were rated unsatisfactory (D or F).

Need to specify the (inferred) reason

```
table(produce_48_data$Second_Eval, produce_48_data$Country,
      dnn=c("Second_Eval","Country"))
```

```
##            Country
## Second_Eval China Japan Korea USA
##           A     0     4    10   0
##           B     0     6    14   0
##           C     1     8    12   1
##           D     1     9     6   0
##           F     0    10    12   0
##           X     0     2     0   0
```

Cross-table: **First__Eval** as row, and **Second__Eval** as column

[71] https://www.adaymag.com/2020/10/22/japanese-vs-korean-idol.html
[72] https://star.ettoday.net/news/1192170
[73] https://star.ettoday.net/news/1181576

```r
table(produce_48_data$First_Eval, produce_48_data$Second_Eval,
      dnn=c("First_Eval","Second_Eval"))
```

```
##           Second_Eval
## First_Eval A B C D F X
##          A 6 3 4 1 1 0
##          B 4 8 5 5 3 0
##          C 3 6 4 3 6 0
##          D 0 3 5 3 4 0
##          F 1 0 4 4 8 2
```

List the names of the six contestants who got A → A.

There are six contestants who got A's in both evaluations.

Miyawaki Sakura (宮脇咲良) is the only Japanese participant, and she was ranked 2nd in the final debut. The other five contestants are all from Korea.

Lee Chae Yeon (李彩演)
Known for her excellent dancing skills (citation needed)
Also made it to the final debut.

Lee Ka Eun (李佳恩)
Explain her actual ranking (5th) and the published ranking (14th) (citation needed)
https://www.ptt.cc/bbs/KoreaStar/M.1605669895.A.97D.html

The remaining three did not do so well in the audience votes. The three contestants are Na Go Eun (羅高恩), Lee Ha Eun (李河恩), and Hwang So Yeon (黃召硯). They still deserve an honorable mention here.

Note that possessing a high talent does not guarantee being selected to debut in the new girl group; having a low starting point does not result in first-round elimination, either.

```r
inds_A_to_A = which(produce_48_data$First_Eval=="A" & produce_48_data$Second_Eval=="A")
produce_48_data[inds_A_to_A, columns_to_show]
```

```
## # A tibble: 6 x 6
##   Name_Chn Name_Eng        DOB        First_Eval Second_Eval Final_Rank
##   <chr>    <chr>           <date>     <chr>      <chr>            <dbl>
## 1 宮脇咲良 Miyawaki Sakura 1998-03-19 A          A                    2
## 2 李彩演   Lee Chae Yeon   2000-01-11 A          A                   12
## 3 李佳恩   Lee Ka Eun      1994-08-20 A          A                   14
## 4 羅高恩   Na Go Eun       1999-09-03 A          A                   29
## 5 李河恩   Lee Ha Eun      2004-10-30 A          A                   48
## 6 黃召硯   Hwang So Yeon   2000-08-21 A          A                   60
```

Other notable participants:

Jo Yuri (曹柔理): A → F

Description here

Yabuki Nako (矢吹奈子): F → A

Description here

Matsui Jurina (松井珠理奈): B → B

Description here

Kang Hye Won (姜惠元): F → F

Nevertheless, Hye Won made it to the debut of the *IZ*ONE* girl group, and she was ranked 8th out of all 12 finalists.

Next step: Breakdown the 1st-cross-2nd table by country (Korea and Japan)

1st-cross-2nd table: Korea

```
produce_48_korea = produce_48_data[which(produce_48_data$Country=="Korea"),]
table(produce_48_korea$First_Eval, produce_48_korea$Second_Eval,
      dnn=c("First_Eval","Second_Eval"))
```

```
##            Second_Eval
## First_Eval A B C D F
##          A 5 2 4 1 1
##          B 3 7 3 2 3
##          C 2 4 4 3 4
##          D 0 1 1 0 2
##          F 0 0 0 0 2
```

1st-cross-2nd table: Japan

```
produce_48_japan = produce_48_data[which(produce_48_data$Country=="Japan"),]
table(produce_48_japan$First_Eval, produce_48_japan$Second_Eval,
      dnn=c("First_Eval","Second_Eval"))
```

```
##            Second_Eval
## First_Eval A B C D F X
##          A 1 1 0 0 0 0
##          B 1 1 0 2 0 0
##          C 1 2 0 0 2 0
##          D 0 2 4 3 2 0
##          F 1 0 4 4 6 2
```

3.3 *Produce 48*: Additional Resources

What happened next?

The final rankings were arbitrarily changed by the show organizers, because they received under-the-table monetary transactions from some entertainment companies, such as Starship and Woollim (Gu, 2020).[74]

What were the political dynamics at play?
The show organizers received gifts from Starship and Woollim.

Evidence: Three contestants were unfairly eliminated from the debut in the final round of the show. (citation needed)

The media announced two of them:

Lee Ka Eun (李佳恩): Actual ranking 5th, published ranking 14th.
https://disp.cc/ptt/KoreaStar/1brmSfeA

Han Cho Won (韓賀瑗): Actual ranking 6th, published ranking 13th.
Cho Won's Wiki: http://bit.ly/45HLq8j

[74] https://www.koreastardaily.com/tc/news/122447

K-pop fans suspected that the third contestant artificially removed was Japanese, who was replaced by another Korean contestant for the *IZ*ONE* debut.[75]

Additional suspects by the K-pop fans.

Many K-pop fans wondered why Jang Won Young (張員瑛) was ranked 1st place in the finals, while the well-known Japanese contestant Miyawaki Sakura (宮脇咲良) came in 2nd. Traditionally the first place (a.k.a. the center position) for the group debut is given to a Korean participant rather than an international one, despite the emerging popularity of K-pop survival shows featuring multicultural performers (Ahn, 2022). Therefore, K-pop fans suspected that Won Young was artificially moved to first place, and that Won Young's actual ranking was between 2nd and 5th.[76] Sakura's official ranking was second place, but she was most likely to be the original first place, i.e., received the most votes in the final round.[77]

The importance of being selected in 1st place for the group debut:
C-position or center position in K-pop (citation needed)

For example, what were the privileges of the first place artists in *Idol School* and *PRODUCE 48*?

Usually the 1st place in the debut get the center position in the newly-formed K-pop group. (citation needed)

The girl group *fromis_9*: Roh Ji Sun (盧知宣) won first place in the finals of *Idol School* reality show, so she got the center position for the *fromis_9*'s debut single song. Ji Sun was also uniquely featured on the debut album cover as a special issue, giving her excellent recognition and visibility.[78]

However, the *IZ*ONE* assigned the center position to Miyawaki Sakura (宮脇咲良), not to the *PRODUCE 48* champion Jang Won Young (張員瑛).[79]

Idea: We can assume that the vote manipulation happened only in the final round of the show.
News citation needed.

Comparison: the 20 participants in the final round vs the rest

Or compare in other rounds: survived vs eliminated ?!

Combine the ratings in the two evaluations:

Average method or take the best score only ?!

4 Tentative Placeholders

Write something here

4.1 Test for Non-English Characters

CJK = Chinese, Japanese, Korean

Chinese example

RStudio 有辦法打中文嗎？

```
print("大家好，很高興能認識你們！")
```

```
## [1] "大家好，很高興能認識你們！"
```

[75] https://www.ptt.cc/bbs/KoreaStar/M.1605679750.A.B79.html
[76] https://www.kagit.kr/posts/123744
[77] https://bit.ly/4n0QuKG
[78] https://bit.ly/473Dcae
[79] https://star.ettoday.net/news/1870619

Japanese example

思い出にするにはまだ早すぎる

```r
print(" みやわき さくら")
```

```
## [1] "みやわき さくら"
```

```r
print(" 宮脇 咲良")
```

```
## [1] "宮脇 咲良"
```

This template does not support Korean characters yet.

4.2 R Markdown Narrative

This is an `R` Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using `R` Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded `R` code chunks within the document. You can embed an `R` code chunk like this:

```r
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

4.3 Including Plots

You can also embed plots, for example in Figure 3:

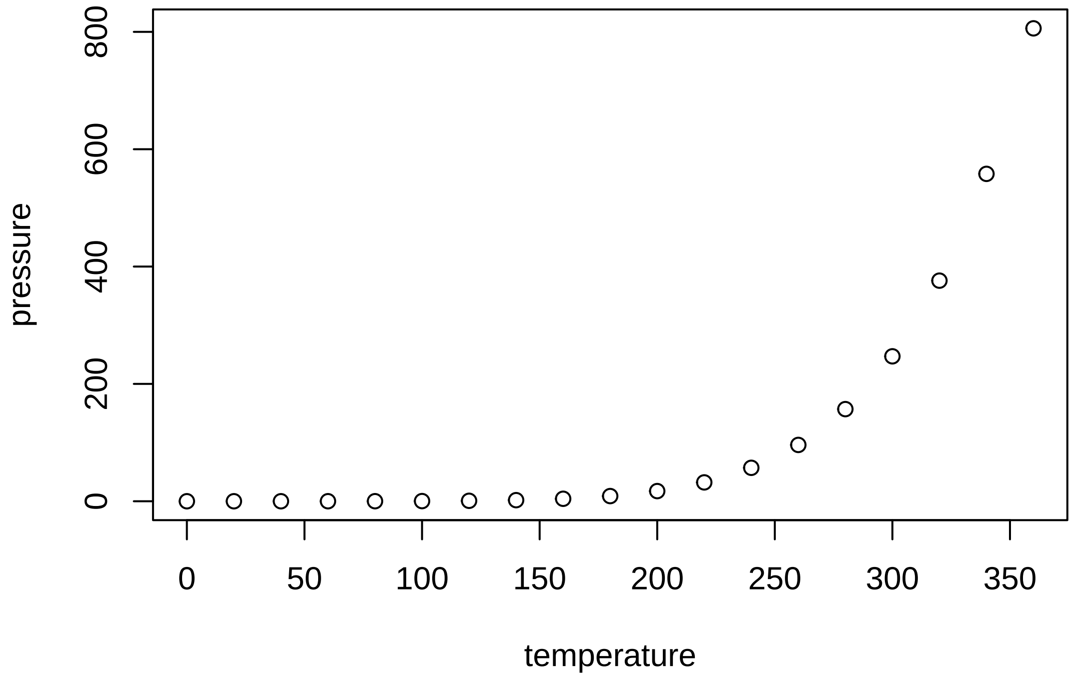

Figure 3: Test Plot

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the `R` code that generated the plot.

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